HELIOS PI HAT - RGB(W) LEVEL SHIFTER PINOUT

SN74AHCT125 Based Level Shifter

open source hardware

Product Information: https://uurnd.com/p/helios-rgbled-pihat

I2C INPUT

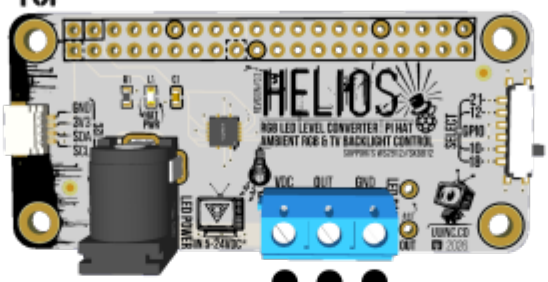
GROUND	GND	1
PI 3V3	3V3	2
I2C	SDA	3
I2C	SCL	4

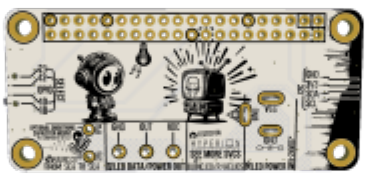
LED DC INPUT

GROUND	GND	5
5-24VDC	LEDPWR	6

PI HEADER

TOP





BACK

GPIO SELECT

X	GPIO	SELECT

LED OUTPUT

7	GND	GROUND
8	OUT	LED DATA
9	VDC	LED PWR

LEGEND

- POWER
- GROUND
- DATA
- SWITCH



LEDPWR Voltages: 5VDC - 24VDC
LEDPWR Max Current:
Barrel Jack: 24V8A - 12V8A - 5V8A
Screw Terminal: 24V18A - 12V18A - 5V18A
Main Chip: SN74AHCT125RGYR
Device Info:
The SN74AHCT125RGYR is a quad buffer/line driver specifically designed to interface between different voltage domains, most commonly acting as a 3.3V to 5V logic level shifter. It is highly regarded for converting 3V microcontroller logic (like Raspberry Pi or Teensy) to 5V device logic (like NeoPixels/WS2812B/SK6812 LEDs). However because the Raspberry PI is powering the SN74AHCT125RGYR, this board supports any compatible RGB(W) LED from 5V to 24V so long as it supports 5V Logic.